OpenMathMap: Interaction

Carmela Acevedo

Computer Science
Jacobs University Bremen
Campus Ring 1
28759 Bremen
Germany

Type: *Guided Research Thesis*
Date: *May 19, 2014*
Supervisor: *Prof. Dr. Michael Kohlhase*

Abstract

Sorting through information can be complicated. Data that is not organized provides no structure for users to make connections with. Semantical search aids users to group results in a meaningful way which facilitates filtering. It is still not clear, however, how to search through this data and how to show results. In order to study new ways to display this hierarchical datasets with semantical meta-data we can use maps. Since mathematics is such a fast growing discipline where an average of 120 thousand papers are published per year we seek to provide a better interface to interact with it. The whole set of mathematical publications provided by the journaling serviceZentralblatt is used to generate a map in which each country is a Mathematical Subject Classification (MSC) class. A platform was built to allow further search and interactive capabilities. Features were developed to enable the use of the map as a tool to selectively obtain results while taking advantage of the semantic and hierarchical properties of the dataset at hand. Finally, this system was evaluated to determine how successful the map is as an intuitive complement for search. The map proved to be a useful interface for interacting with and understanding the results that were being displayed.

Contents

1 Introduction **3**

2 State of the art **5**
2.1 Math Atlas 5
2.2 Paperscape 6
2.3 OpenMathMap 6

3 Preliminaries **8**
3.1 Zentralblatt 8
3.2 Map Services 8

4 Scenario and Use Cases **9**

5 Use cases implementation **11**
5.1 Preliminary and general features 11
5.2 Current Areas of Mathematics 12
5.3 Result filtering 13
5.4 Relevant author's information 14

6 User Interface **15**
6.1 General components 15
6.2 Current Areas of Mathematics 16
6.3 Result filtering 17
6.4 Relevant author's information 17

7 Evaluation **19**
7.1 Methods 19
7.2 Results 20
7.3 Discussion 21

8 Conclusions and Future Work **23**

1 Introduction

We currently live in the times of big data. Data management has been revolutionized in the recent years. Nowadays, servers are no longer confined to rooms filled with racks with air conditioning, but there are enormous data warehouses serving information all over the world. Considering the rate of increase in information storage, it has become a great challenge for humans to discern information that interests them. Currently the most common tool for going through a large amount of data are search engines. But, first of all, it is difficult for the user to compile a query if she does not have an understanding of the structure of the database she is presented with. Secondly, it is challenging to display the results in a helpful way. We are currently in the same state as a supermarket that ranks its products according to how popular they are. But the users would have more incentive to investigate different results if they knew how the results are related to one another. Moreover, grouping the results by a common topic could guide the users to interpret them and provide an overview of the underlying structure of the information.

Organizing search results brings two challenges to the surface. One of them is how to organize objects hierarchically in a way that provides a better overview of the entire set of search than by listing them individually. The second challenge is to visualize and display this information so that it is intuitive enough to guide the user to discover new connections between results by diving into a topic of interest. Maps could be a potential answer to these problems. People are familiar with maps from a young age. We learn how to read and navigate them. They display the geographical world in a global, hierarchical way and at the same time help us comprehend our surroundings. Having global and local information simultaneously provides the information context. The topic of this guided research is to investigate whether a map can provide a hierarchical overview of non-geographical information that would promote discovery and investigation.

During the last fifteen years there has been an average of more than 120 thousand publications in mathematics per year [Zen13a]. The last version of the Mathematical Subject Classification (MSC)[MSC12] has been decided upon jointly between Mathematical Reviews (MR) and Zentralblatt für Mathematik (Zbl). All publications have been assigned a main MSC class they belong to and a number of secondary classes related to it. With this information a map of mathematics was generated. This is a good opportunity to research the possibility of using this map to retrieve information related to these publications. This hierarchical map contains all first and second level MSC classes [DKL13] [DK13] [DÏ3]. Classes have sizes proportional to how many publications they have. Most importantly, the distance between every two classes is inversely proportional to how similar they are. We seek to take advantage of the fact that two classes that are close to each other contain publications that are related. This platform could not only help us obtain more information about publications but also about authors and whole areas of mathematics.

This project will seek to investigate the use of the map as a tool for a semantic approach to search. To illustrate this we consider the scenario of a student that is undecided as to what to focus on later on in her career in mathematics. She wants to be able to narrow down her

choices to a handful. For this she needs information about which areas are thriving at the moment, what they are about and which authors are active in them. We will evaluate how integrating interactive features to the map facilitate information retrieval in order to answer these specific questions. Finally, we prove that by using maps for search the user gains more information about the global topic and at the same time finds answer more easily.

2 State of the art

There have been attempts to popularize semantic search engines [GC06]. Improvements in technology have made it possible to store very large amounts of data. During the last few years numerous frameworks such as D3.js in 2011 [Wik14a] and the Google Chart API in 2007 [Wik14b] have surfaced to try to visualize this information. They allow for simple and elegant data visualization on the web. There are map interfaces that simplify the navigation of online maps such as Google Maps [Goo14] and Nokia's HERE maps [Nok14]. These frameworks also offer the possibility of visualizing data, such as voting results for example, in map form, since this is an easier way for humans to get a grasp of the information.

2.1 Math Atlas

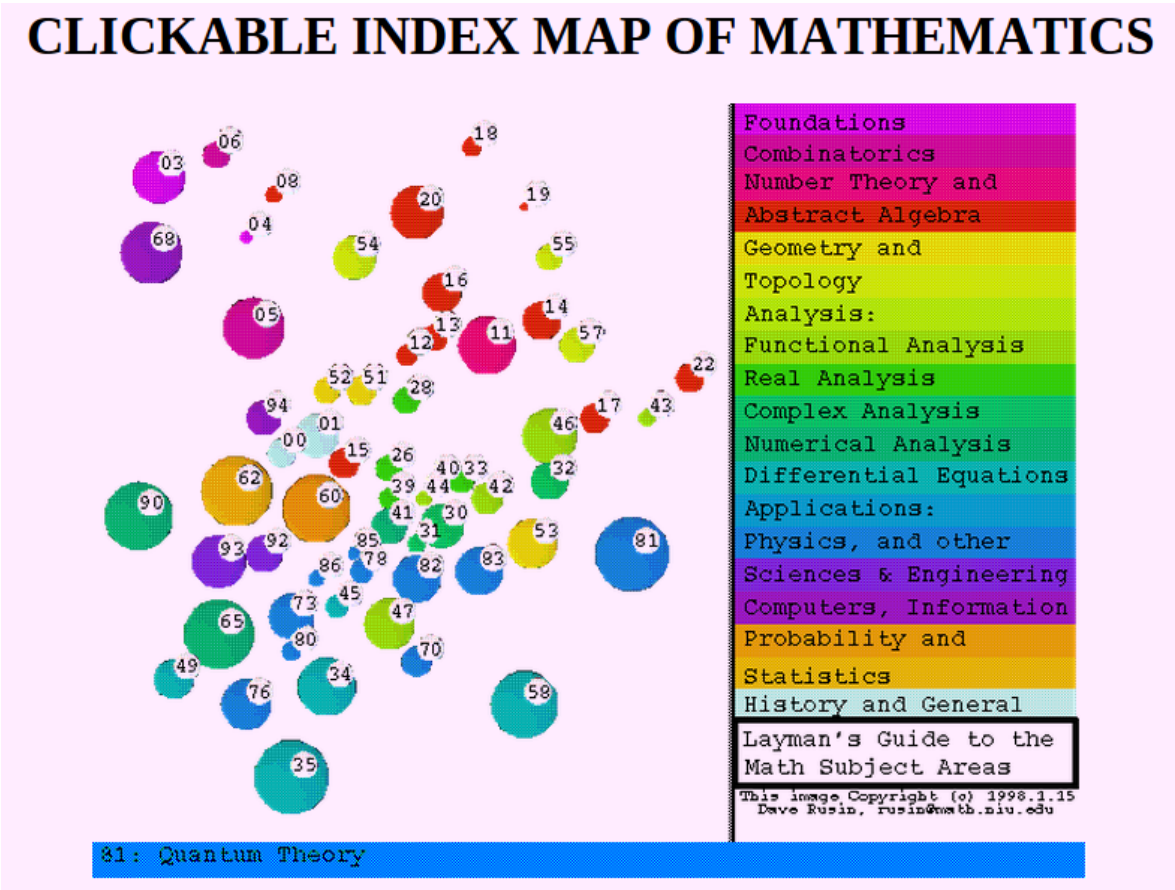


Figure 1: Rusin's Math Atlas [Rus01]

Long before these technologies surfaced Dave Rusin created a Math Atlas in the year 2000 [Rus02]. This atlas shows a circle for each of Zentralblatt's Mathematical Subject Classification (MSC) classes [MAT14]. These have a size depending on the amount of papers published in them. They are positioned according to their relation to the rest of the other categories (Fig. 1). The distance between classes is directly related to how many papers have been pub-

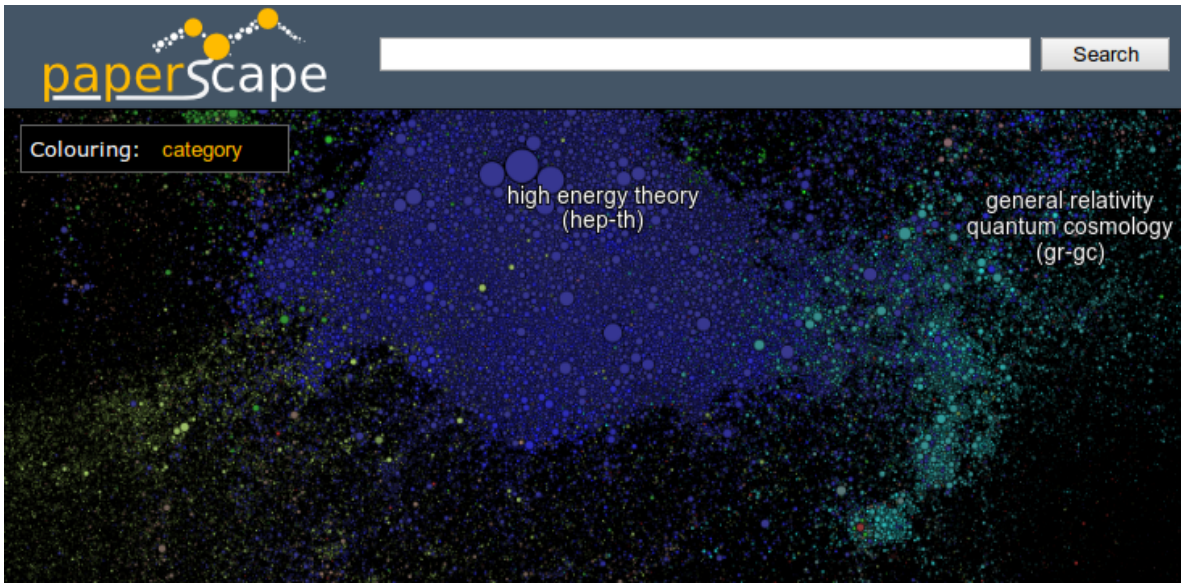


Figure 2: Paperscape

lished under both categories. This is an almost purely visual representation of mathematics. The only possible interaction with the map is clicking each circle which redirects the user to another page describing the corresponding MSC class.

2.2 Paperscape

Paperscape is a similar initiative where each circle represents one paper from the arXiv [GK14a]. Again the distance between them is related to how similar they are, but in this case the size is directly related to how many citations each specific publication has. This map is updated daily by obtaining all the arXiv's publications, rendering the map to a bitmap and displaying it in a canvas HTML5 object [GK14b]. This project is a lot more interactive than the previous one. Here users can search for papers by several criteria like author or keyword highlighted in the map. It is also possible to click individual papers and obtain more information about them (Fig. 2). The creators are continuously working on integrating other features.

2.3 OpenMathMap

In this project mathematical publications provided by Zentralblatt were used to create a multidimensional space according to which MSC class they belonged to. Multidimensional scaling was later used to map this space into a two dimensional structure. MSC classes are subdivided into smaller classes, the same process was followed with these to create maps inside each "country" that was their parent MSC. As in Rusin's map each MSC has a size corresponding to the amount of papers published inside it and their closeness represents the ratio of papers that are published in both MSCs simultaneously (Fig. 3).

At this point OpenMathMap is at a similar level of interactivity as Rusin's map of math. When a user clicks on an MSC in the map a pop-up shows what the name of this specific area is and offers links to PlanetMath and Zentralblatt for deeper descriptions of it.

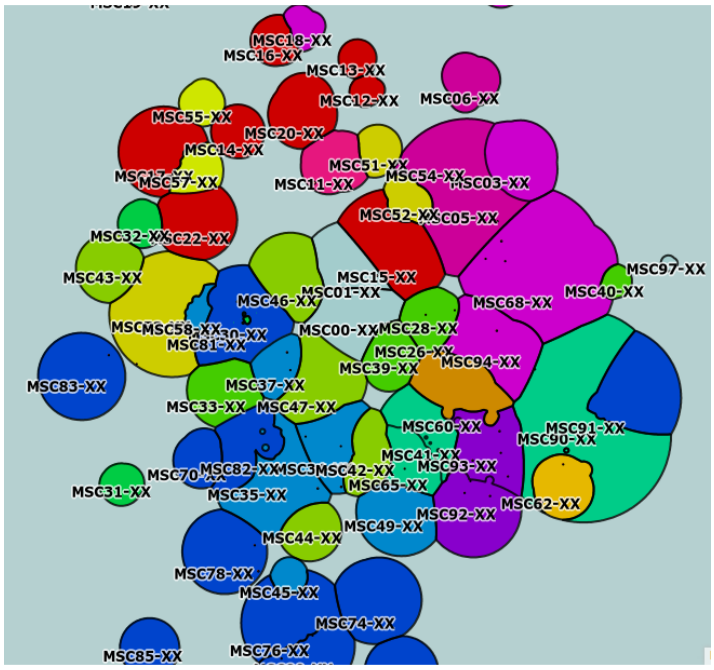


Figure 3: Open Math Map

3 Preliminaries

In the scope of this project we are dealing with mathematical publications. All the data was provided by Zentralblatt and it was later put together into one cohesive system with the aid of mapping tools.

3.1 Zentralblatt

Zentralblatt is a service that provides reviews and publication content from pure and applied mathematics [Wik13] [Zen13b]. It contains a comprehensive database of all mathematical publications dating from 1868 which contains more than 3 million bibliographical entries. It provides several ways to search this database. Some examples are: searching by author, journal or a formula. In order to keep an organized structure of these publications they have decided upon a classification jointly with the Americans Mathematical Society's Mathematical Review. This Mathematics Subject Classification (MSC) counts with three nested levels. Each published paper is given a primary MSC class and several accompanying secondary classes.

The search results provided in Zentralblatt's portal are always accompanied by their corresponding class in order to provide further information for the user. There is also a way to browse this classification top down if one wants to find results under a specific class. However, This interface is lacking the opportunity to give the user an understanding of how these search results are related to each other. We are seeking to provide an interface that will give a visualization of such results. In the scope of this project we will focus on author search, MSC class search and formula search. The latter searches for formulas in LaTeX format and text that appear in publication abstracts.

3.2 Map Services

To serve a map in the web there are certain frameworks that need to be put together. A file, an XML file for example, is created that contains all the map information. This file is later used to generate tiles that contain the images of the map in different levels. It is also parsed and uploaded into a database to serve the geometrical information of the map to the users on demand. There is also a service in charge of displaying the tiles of the map in the order they are supposed to be in. On top of this then there are built-in services that can be integrated into the system taking advantage of the underlying infrastructure.

4 Scenario and Use Cases

We will focus our research on one specific scenario from which we develop three use cases. The goal is to evaluate whether a map is a viable search tool. Because of this we have to set up a scenario that can be performed with existing techniques in order to establish a comparison.

Scenario

We consider a student, Stephanie. She is going to soon finish her undergraduate studies in Mathematics, but is not sure in what to continue her studies. She has been accepted into five different master programs. There are several aspects that influence this decision. She has several areas of interest but she wants to be certain that she does not go for an area of mathematics that is not thriving at the moment. Another important factor in this decision are the professors and students that she could later work with. This is relevant for deciding to what institution she would rather go to. She already has some topics of interest in her mind but she has not been able to find too much information. When she tries to search for this she gets results that are too varied and it is really hard to realize how biased what she is reading actually is. She would like to be able to have some data that is objective and that somewhat sheds some light on her problem.

From this scenario we extract three use cases that we will be focusing on:

Current Areas of Mathematics

She wants to give an overview of what has been worked on recently in mathematics. It is important for her to know what areas of mathematics seem to be making progress at the moment. Whenever she searches for this she obtains independent results for data that is very focused on specific topics. She has to put all this information together in order to compare it. Nevertheless, she has not been able to get an overview of the whole situation.

The OpenMathMap provides the backbone to display all of this information at once. We will solve this problem by providing an overlay to the map that shows how active areas of mathematics currently are.

Relevant author's information

She has found out who the professors are that work in the places she is admitted in but she is not sure what they are actually publishing in. Furthermore, she would like to know if what they are publishing on is consistent or if they have not yet committed to a special area.

The map can be used to provide this information in a similar way as before. By providing an overlay to the map that displays this information she could look at it all at once and analyze it better.

Paper search by formula, text and class

Since she is aware of some topics that she is interested in she would like to be able to search for these and find publications in the classes that are possible candidates.

The results of a search can be displayed on the map to show a general distribution of them. Knowing the geographical location of the classes can further help to locate a subset of results that are of special interest.

5 Use cases implementation

In order to reach a platform that provides the mentioned use cases we have to start by developing some basic services.

5.1 Preliminary and general features

We started with a tile map being served by a Javascript framework [Lea14]. We obtain the OpenStreetMap (OSM) [OSM14] XML file that was used to generate the map. This file contained all the relevant geometrical information like the paths that define the countries. This file was later uploaded to a geometrical database (PostGIS) [Pos14]. This has the advantage that checking if a point is inside of a country is very simple and we can also provide extra information about countries on demand like its centroid, area or neighbors. At the current state of the system it is enough to generate a new OSM file and upload it to the database in order to reuse it for a newly generated map.

After having this available and being served by a framework (Django in this case) we create navigation opportunities for the user. These come in two forms, by exploring and searching. Exploring means the ability of clicking on the map and obtaining information. The user can zoom in and out and select a country to know which one it is. This is done by obtaining the click position on the map and finding which country this point belongs to. The current state of the map contains countries inside other countries in which case we had to check what country was the wrapping one in order to decide where this point belongs. Searching was implemented by providing an input text bar for the user. This bar handles four types of search:

- Author search by name
- Class search by name or number
- Formula and/or text in abstract search
- Free search which provides the results of all the previous ones.

Search results from all these modes result in markers appearing on the map. Each marker when clicked launches a popup with the title of the marker and a link to the Zentralblatt page displaying the result. These markers are positioned in the centroid of the class they belong to which in the current state of the map always lies inside of the country. The country that the marker belongs to is decided differently for each mode. An author marker is placed in the MSC class that this author has published the most in. A class marker is put on the corresponding class. A formula/text marker is put on all the classes this paper was published under.

The information provided by the search is retrieved in different ways as well. For the author search we used a file containing all the publications that Zentralblatt has available. This file contains around 3 million entries each detailing publication year, title, authors, classes among others. We processed this file and created author instances in our database containing the amount of publications and average year under each class. For the class search we use the

Zentralblatt classes with the data available in the OSM file. For the formula search we use a service provided by Zentralblatt which allows for the search of formula's in LaTeX format and text in abstracts or titles of publications.

5.2 Current Areas of Mathematics

A solution to this problem is to generate an overlay of the map that provides relevant time information about MSC classes. By doing this all the information could be displayed to the user at once. This would facilitate a better general understanding and first encounter with the subject manner. This is done by retrieving the boundary of each country from the database and creating a layer containing all of these polygons.

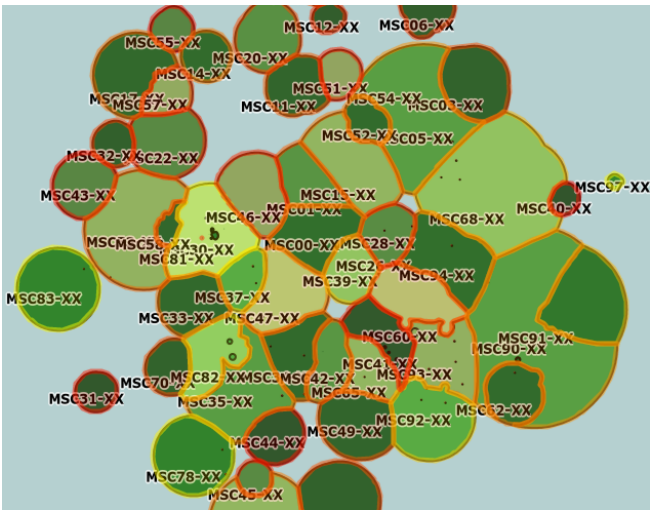


Figure 4: Average overlay.

We provide the service a color which currently is generated to be between yellow (`'ffff00'`) and red (`'ff000'`) mapped between 0 and 100 respectively. The number that is provided for each class is something that we had to investigate. We have the publication year of all the papers available in Zentralblatt, since these are 3 million papers it was not feasible to store all this information on the database in the scope of this project. We did a preprocessing step where we obtained only years of publication and class of publication. With this information we generated statistics for each class. It was necessary to narrow down the information to a single number.

We have to decide on what is what we want to display with such overlay. This is which areas of mathematics are currently the most popular in terms of publication. For this we have to pick a value that is related to the change in publication behavior over time. At first we picked the average year of publication per class. This however had the problem that classes with very different behavior had the same average (Fig. 4). Then we tried to look at how the maximum and minimum year of publication would behave.

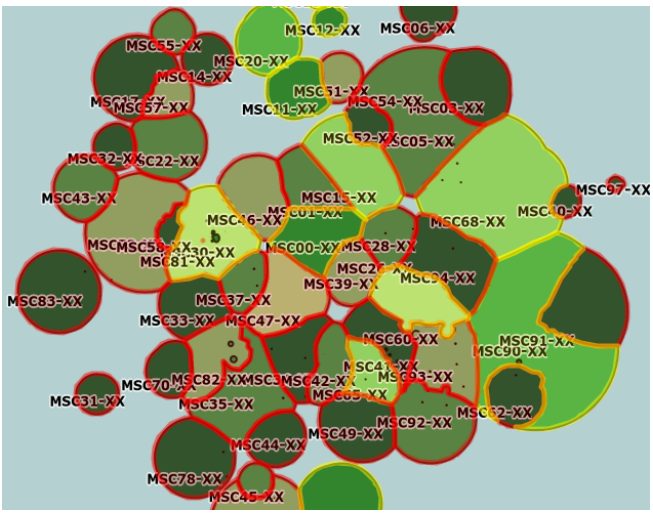


Figure 5: Last year overlay.

The maximum has the problem that
most classes have a very recent publication and hence all of them had a value in the last five years (Fig. 5). The minimum did not offer any information on the current behavior of the respective class.

Finally we decided on comparing the amount of publications that have occurred in a class during the last 5 years with the rest of the publications (Fig. 6).

For this we count the amount of publications in a class and calculate the percentage of these that have been published in the last 5 years. This offers an insight on how active this class is compared to the past. Young classes like education and relativity have very high values like 44% and 19% while other classes only have 5%. It is interesting to see that all classes with high values are ones that are islands. This could be because they are young classes and have not had enough publications in order to be considered very similar to others.

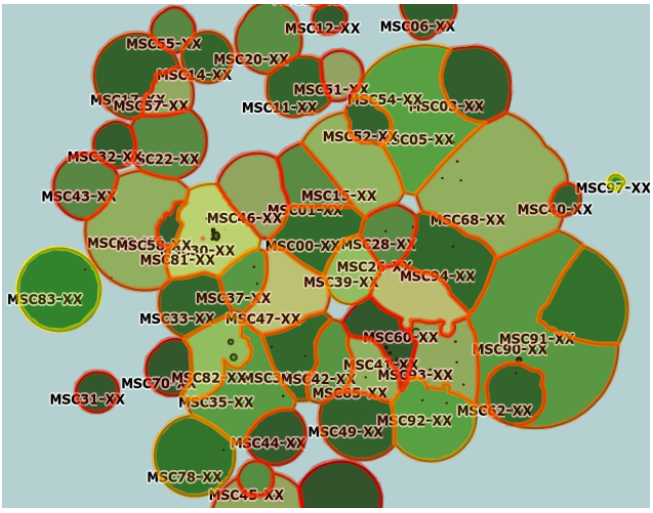


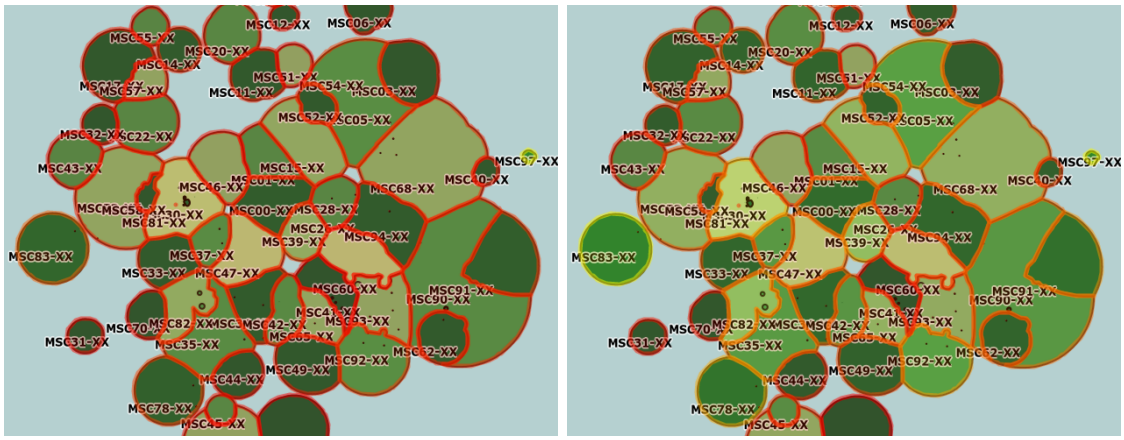
Figure 6: Last 5 years overlay.

A problem that was encountered with all of these measurements was the presence of outliers. Education is generally an outlier because it is a very young class and hence it always has very different values to the rest.

This is relevant since we scale all values between the minimum and the maximum in order to set the gradient color values. We solve this issue by taking the maximum of the values in between a certain range, if they are an outlier we do not take them as a maximum (Fig 7. This leads to the values being scaled in a meaningful way and outliers obtaining the same color as either the minimum or the maximum which is still valid.

5.3 Result filtering

When searching results appear all over the map. Our user is only interested in her top MSC classes and hence would like to filter the results. Since there are potentially a lot of results a solution that lets the user pick markers individually is not viable. We also want to take advantage of the fact that similar classes are geographically near in the map. A drawing selection tool could solve this problem. After obtaining the results a tool can be offered. This tool will let the user draw on top of the map. This will generate a layer with geometrical objects that encapsulate the desired results. After a subset of the results has been selected they can confirm their selection and only the selected ones would remain on the map. This



(a) Normalizing with maximum

(b) Normalizing without outliers

Figure 7: Normalizing values considering outliers.

is done by going through all the markers and checking if they are inside one of the drawn polygons. Since the map can only display a limited amount of information we offer a link to the Zentralblatt site with information about all the individual results. This link is updated throughout the filtering process to refer only to the relevant markers.

5.4 Relevant author's information

A single marker symbolizing an author is not enough to encode their trajectory throughout time. We are already positioning the marker on the class for which the author has contributed the most. This does not offer information about what other classes this author has published in or the time development of his career. For this purpose markers of authors have a popup that includes two buttons, "Classes" and "Timeline". Both these buttons generate an overlay over all the countries representing the MSC classes where this author has published.

The "Classes" button is meant to provide an understanding of which classes this author has published in. For this we create polygons on top of each class he has contributed to. We color each polygon according to the amount of papers he published under the respective category. We use yellow for the maximum amount of publications and red for the minimum taking advantage of the fact that yellow is brighter. Generally authors have a specific area of focus which results in overlays with one or two yellow classes and the other classes they have explored with a red polygon.

The "Timeline" button seeks to offer a timeline for the author's work. This is similar to the first use case where we try to display the progress of the classes over time. This case is simpler since most authors have less then 100 publications in an average of 10 classes over half a century. In this case the average does offer a meaningful intuition of the authors publication behavior. Commonly there also are no outliers because authors publish during their active careers which is a short window of time.

6 User Interface

In order to have all of these features we need an interface for the user to interact with the map. There are several components that need to be created to support all of the use cases. They are afterwards integrated to provide all of the desired interaction.

6.1 General components

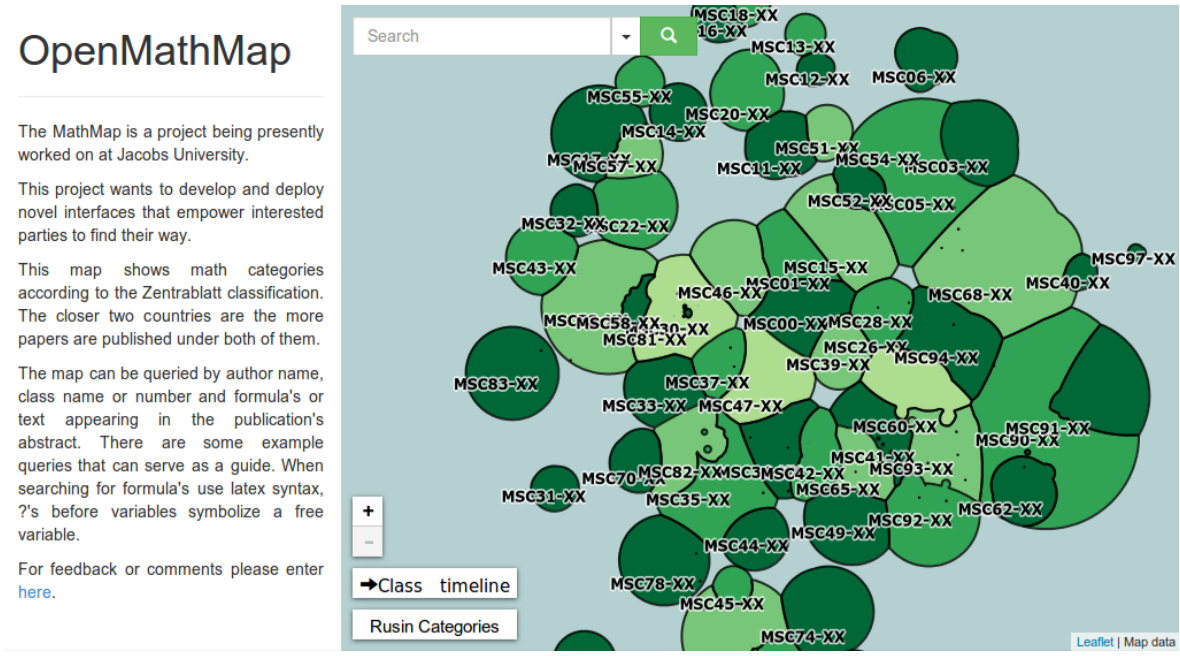
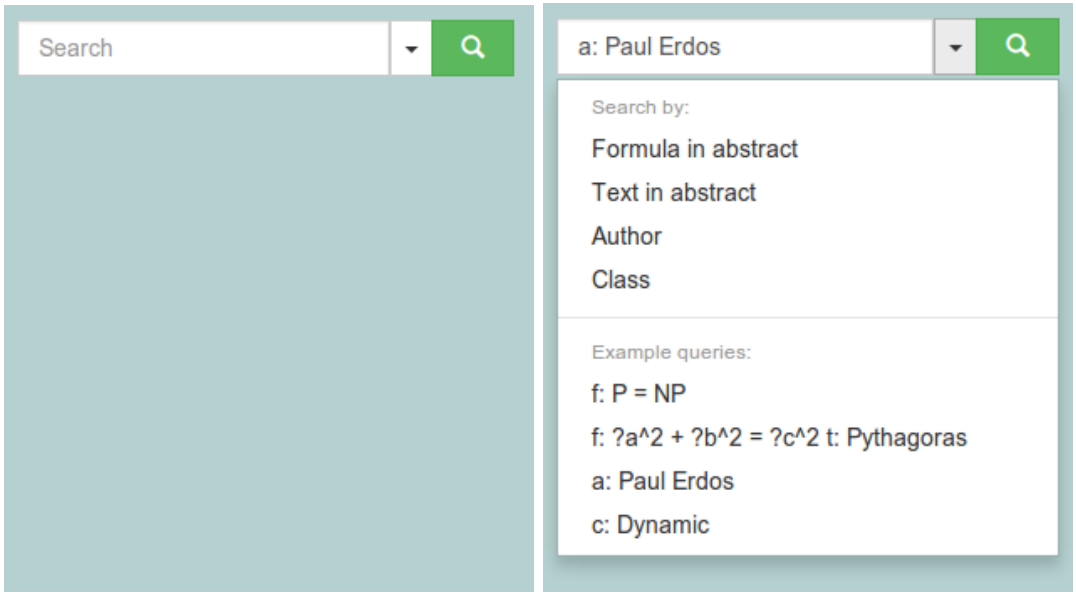


Figure 8: General layout.

The map is displayed by the Leaflet service. We have embedded a map instance in an HTML document leaving a column to the left available for information. The Leaflet map allows for zooming along already generated tiles of the map. We have included zooming buttons provided by Leaflet.

A search bar has been placed on the top left corner of the actual map area. This integrates it and informs the user that the search results will be in the scope of the map. This bar contains a text area, a dropdown and a button (Fig. 9a). The text area contains the text that the user types or selects from the dropdown. The dropdown has two functions, select a mode for the search or select an example query. Without any mode the search bar will search on all types of search possibilities which can lead to a cluttered map. For this we provide different modes (Fig. 9b). A mode is selected by placing a shortcut followed by a ":" in front of the query. For example "a: Carmela Acevedo" will search for an author named "Carmela Acevedo".

The results are displayed as markers on the map. For this we use the default Leaflet marker (Fig. 10a). Since it often occurs that markers overlap we are using a service to solve this



(a) Search bar initial state. (b) Content and dropdown

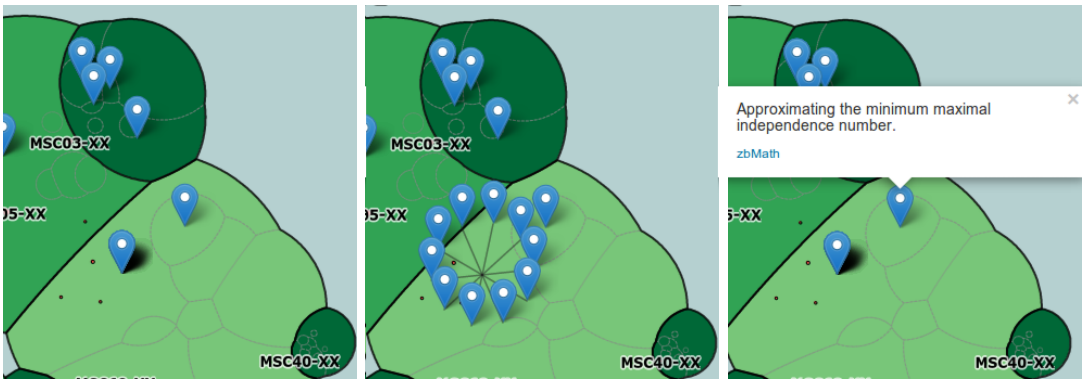
Figure 9: Search bar description.

problem. This service lets markers overlap but when their position is clicked then they are all organized in a circle around their original position. In this way markers can maintain their position while the user has the ability to navigate them all (Fig. 10b). We also add a popup to each marker providing more information about it. All of them have the title of the respective object and a link to Zentralblatt with the specific results data (Fig. 10c). There exists a rectangular area providing extra information about the results. Currently it just hosts a link for all the current markers to a Zentralblatt result page displaying all of them.

We have also included a spinning icon that appears when the system is calculating and is not ready to display any information.

6.2 Current Areas of Mathematics

For this use case we provide a button "Class timeline" which generates the country overlay (Fig. 8). Additionally we use the left information area to provide more data. Below the search there is a HTML div containing a legend for the overlay. We divide values in five equal intervals and assign them a color. Even though there may be intermediate colors in the map this serves as a guideline (Fig. 11a). Whenever a user hovers over a country this HTML div is updated to contain the title of the respective MSC class, the total amount of publications under such class and its respective value (Fig. 11b). This helps to better understand the colors and provides exact information to the user.



(a) Markers. (b) Marker expansion. (c) Marker popup.

Figure 10: Marker overview.

6.3 Result filtering

When a search that requires filtering is performed we add an extra toolbar to the map below the information area. It contains four buttons: draw, edit, remove and confirm (Fig. 12).

Button	Function	Active when:
Draw	Draws a circle	Always
Edit	Edit existing circles	circles exist
Remove	Remove existing circles	circles exist
Confirm	Remove all markers outside circles and circles	circles exist

Additionally there are hover states and popups to guide the user in the process.

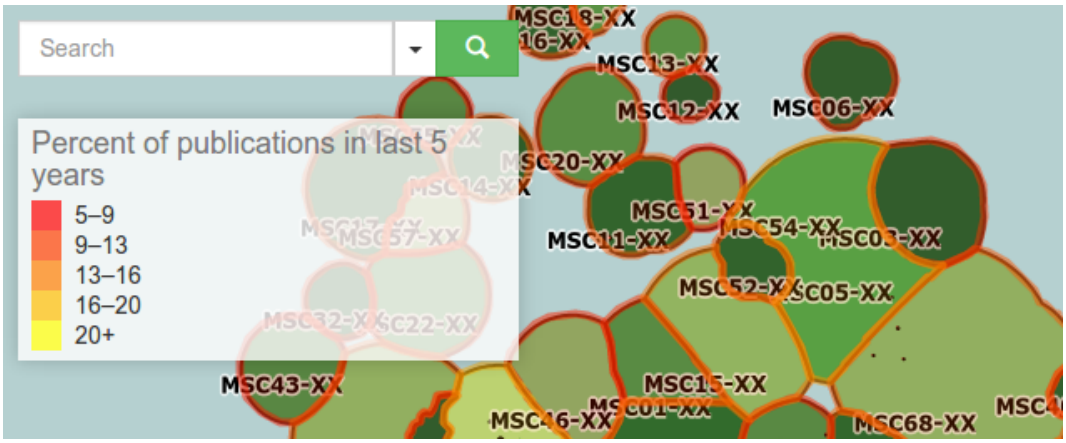
An example workflow would be (Fig. 13):

1. Search for a formula
2. Click draw button
3. Draw a circle
4. Edit/remove/add more circles (optional)
5. Confirm
6. Click results link

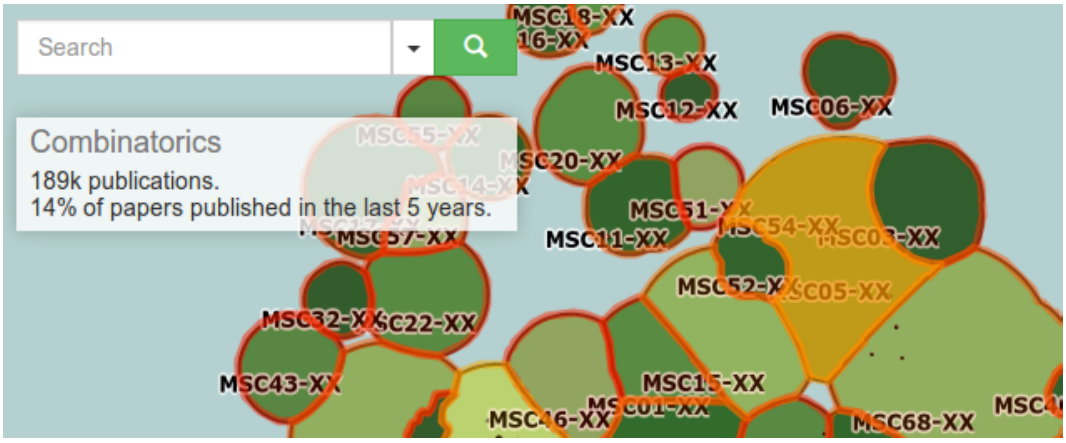
The current drawing tool that is being used has the drawback that it does not allow for simple editing or removing functionality. The user has to click save or cancel after every iteration which could become an irritating for the user.

6.4 Relevant author's information

Since authors can have their own overlays we place two buttons in each of their popups. Each button prompts the appearance of an overlay. This is done in the exact same way as in the



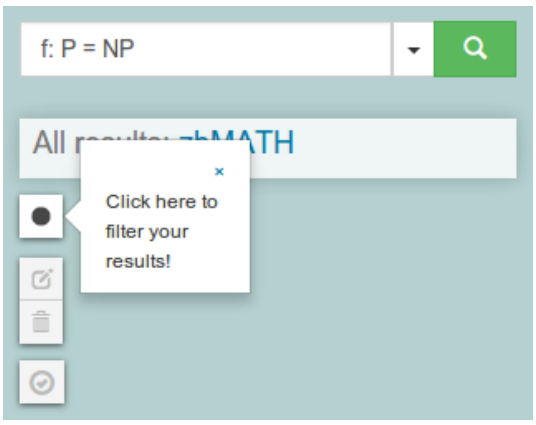
(a) Class timeline with legend



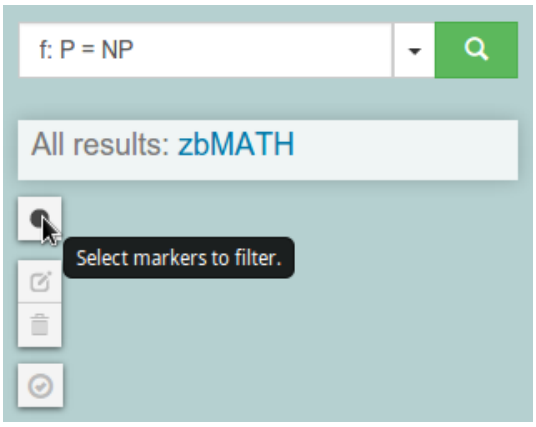
(b) Legend for specific class on hover.

Figure 11: Timeline.

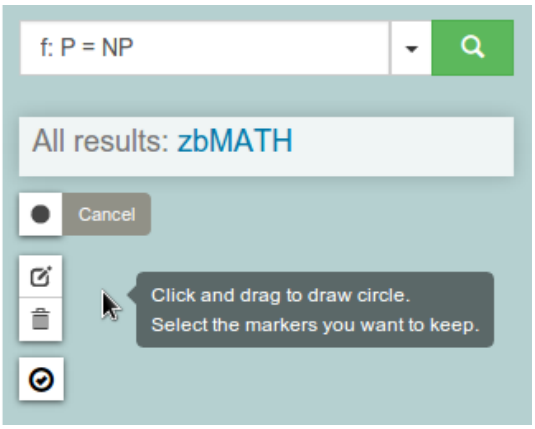
general case. There is also a HTML div displaying special information. In this case it displays the name of the author, total number of publications and a legend for the class version and just the name and legend for the time one.



(a) Selection tool.



(b) Draw button on hover.



(c) Arrow in drawing mode.

Figure 12: Drawing toolbar.

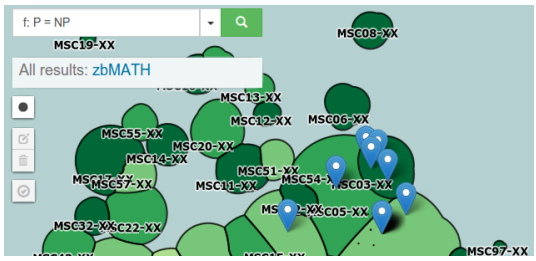
7 Evaluation

In order to determine if this system is a successful aid for searching an evaluation was performed. This process sought to compare the present solution to previously existing systems.

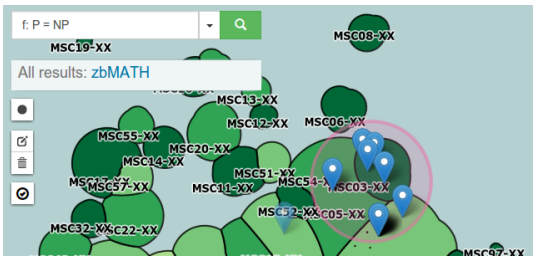
7.1 Methods

Considering the scenario that we have established we chose to evaluate the system on University students in the process of obtaining either an undergraduate or graduate degree. These chosen students have some inclination towards mathematics but are not necessarily mathematics students. There were five students that took part in this process. Each participant was invited to the same setup. They were first presented with the web application and asked to discover what it was. During this time they were encouraged to click, move and type to their liking. After they were familiar with the system they were prompted to perform three tasks only with the map:

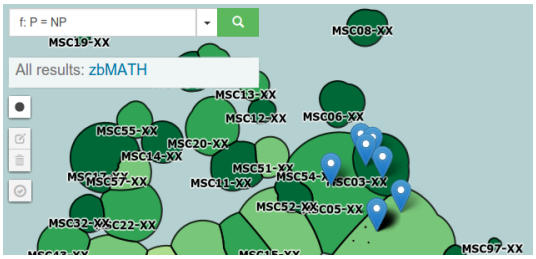
1. Find information related to which class has been the most active in recent years.



(a) Map with results.



(b) Drawn circle.



(c) Filtered markers.

Figure 13: Filter workflow.

2. Find a specific author and describe their career according to their publications.
3. Search for a formula and filter the results to the ones in one specific class.

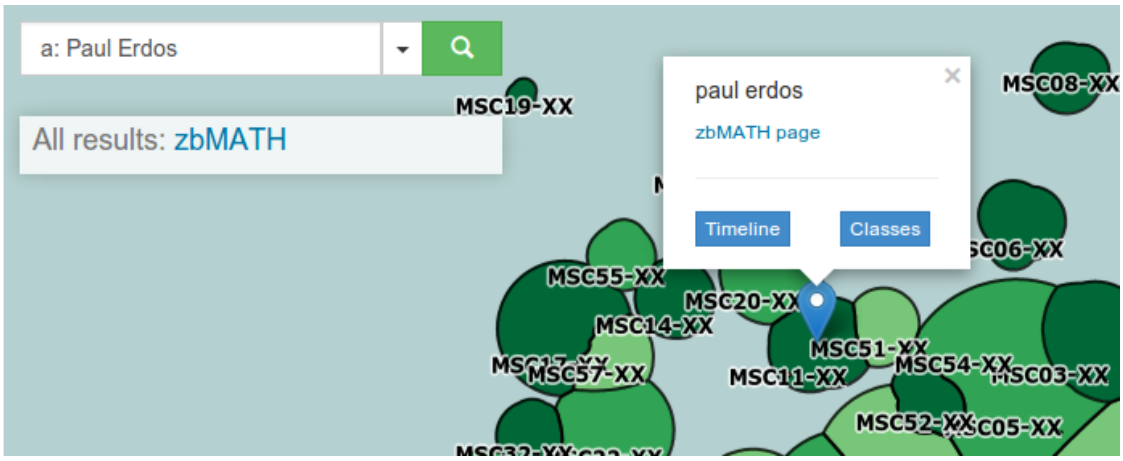
Once these tasks were over they were asked to perform the same tasks in the official search of Zentralblatt. After this supervised part they filled out a survey comparing the ease for obtaining information in both systems (See Appendix).

7.2 Results

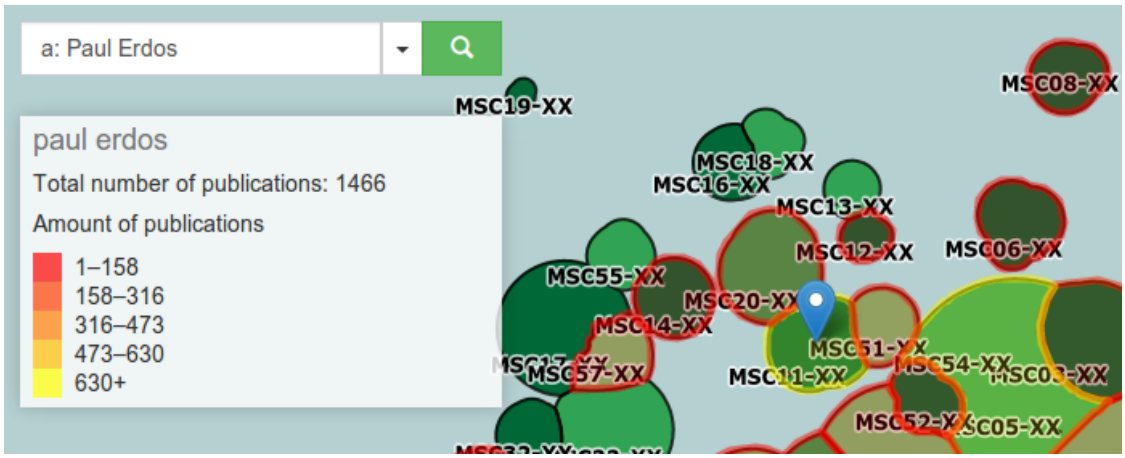
All the participants grasped the concept of the map during the beginning of the evaluation and worked diligently through the tasks. They found the map to be a useful metaphor to display the information with. After having the results and data available on the map they were able to successfully analyze it and understand how results were related to each other.

Participants considered that the overlays that were providing extra information were very useful. Filtering results and finding a specific author was rated to be relatively easy on the map since these tasks are nearly impossible or very hard in the Zentralblatt search. Users did have technical problems with specific tools and buttons that hindered at moments their possibility of obtaining results.

In the first task it was very hard for them to find the "Class Timeline" button. They would attempt to analyze the map in the state it was currently in, trying to map the colors to a time related meaning. Once they found the button it was obvious to them that they could obtain the information they wanted through this button.



(a) Author popup.



(b) Author classes.

Figure 14: Author information.

Another problematic tool was the filtering tool. There are currently four buttons related to this tool. They are very small and have no text. We attempted to make them more informative by offering popups and hover texts for them. This seems to have been unsuccessful. The participants were able to understand the drawing tool but once they had a circle drawn encapsulating some markers they would not find the confirm button. This could be related to the fact that once a circle is drawn the markers outside of it become more transparent and hence participants could believe that the process was over. The editing and removing tools were also troubling because they expect users to confirm after making changes.

7.3 Discussion

There were some technical problems related to the implementation of the system that could be done differently or improved.

The fact that users could not see the timeline button was probably related to the fact that the map is concentrated around some area which they were focusing on. This means that everything else was outside of their field of view and they would not consider. Participants were trying to give meaning to the colors of the map instead. This fact makes it clear that using colors to encode the time is a useful model. Instead of four-coloring the map with different shades of green we could start with the timeline coloring from the beginning. Otherwise we could not use shades of green for the coloring of the map but different milder colors and reposition the button to attempt to make it more visible.

The filtering tool could be improved to be easier to control. We could safely assume that the markers are filtered once the circles are drawn. The outside markers have a higher transparency and we could update the link to contain only the relevant objects. This would avoid the need of having to confirm the selection. This way the selection would also be easier to undo, just by removing the circles that are on the map. The removing and editing tools can also be improved. We could offer a menu on each circle on right click, this is a convention that we could take advantage of.

From the interaction participants had with the map and the answers they provided to the survey we can consider that the use of the map to display the information was successful. Users were able to understand it and moreover they were able to draw conclusions from it and data related to it. Objects that were displayed in the map were quickly interpreted by the participants. This shows us how useful the map can be in hierarchical search scenarios where results have a semantical relation.

8 Conclusions and Future Work

There are certain features that could further improve the interaction with the map. On the interaction side: The current search capabilities of the system are limited to satisfy our three use cases. Queries should be able to be combined in order to provide more specific results. It would also be interesting to be able to compare results between the same type of query. The map is based on the fact that it provides an overview so comparing results would be a logical step to take. On the mapping side: We are currently not fully taking advantage of the hierarchical nature of the map. A new map could be generated that in fact displays individual objects. Hence, users could zoom into the map in areas they are interested in and properly discover with no previous knowledge of the subject. The map could also be integrated with previous search systems, such as the Zentralblatt search, to complement each other. Finally, new maps should be used with the existing framework in order to evaluate whether the results are related to the map or the interactive framework component.

In this project we have investigated the potential of maps as search tools for large hierarchical information databases. For this we developed a framework that enables users to interactively perform three search tasks: obtain general information about the current developments in mathematical publications, find fields of interest of particular authors and filter results according to their semantical relations. We have evaluated this framework on five users and concluded that our tool is more helpful to answer these questions than typical search engines. Therefore we prove that maps are a useful search tool because they provide an overview of a topic while letting users investigate areas without losing track of the big picture. The best tools to facilitate hierarchical interactive search are yet to be found.

Acknowledgements

I would like to thank my supervisor, Prof. Dr. Michael Kohlhase, for his continuous support and advise. In addition I would like to thank the KWARC research group for their help in numerous technical matters.

References

[DÏ3] Jan Wilken Dörrie. "OpenMathMap: Acessing Math via Interactive Maps". In: (2013).

[DK13] Jan Wilken Dörrie and Michael Kohlhase. *OpenMathMap: Accessing Math via Interactive Maps*. 2013.

[DKL13] Jan Wilken Dörrie, Michael Kohlhase, and Lars Linsen. "OpenMathMap: Accessing Math via Interactive Maps". In: (2013).

[GC06] Vladimir Geroimenko and Chaomei Chen. *Visualizing the Semantic Web - XML-based Internet and Information Visualization*. Berlin, Heidelberg: Springer, 2006. ISBN: 978-1-846-28290-4.

[GK14a] Damien George and Rob Knegjens. *Paperscape*. 2014. URL: http://www.paperscape.org/ (visited on Jan. 25, 2014).

[GK14b] Damien George and Rob Knegjens. *Paperscape*. 2014. URL: http://blog.paperscape.org/?page_id=35 (visited on Jan. 25, 2014).

[Goo14] Google. *Google Lat Long*. 2014. URL: http://google-latlong.blogspot.de/ (visited on Feb. 9, 2014).

[Lea14] Leaflet. *An Open-Source JavaScript Library for Mobile-Friendly Interactive Maps by CloudMade*. 2014. URL: http://leaflet.cloudmade.com/ (visited on May 9, 2014).

[MAT14] Zentralblatt MATH. *Zentralblatt MATH*. 2014. URL: http://www.zentralblatt-math.org/msc/ (visited on Jan. 25, 2014).

[MSC12] American Mathematical Society. *MSC2010*. 2012. URL: http://www.ams.org/mathscinet/msc/pdfs/classifications2010.pdf (visited on Dec. 7, 2012).

[Nok14] Nokia. *HERE*. 2014. URL: http://conversations.nokia.com/category/here/ (visited on Feb. 9, 2014).

[OSM14] OpenStreetMap. *OSM XML*. 2014. URL: http://wiki.openstreetmap.org/wiki/OSM_XML (visited on May 9, 2014).

[Pos14] PostGIS. *Spatial and Geographic objects for PostgreSQL*. 2014. URL: http://postgis.net/ (visited on May 9, 2014).

[Rus01] Dave Rusin. *About the MathMap image*. 2001. URL: http://www.math.niu.edu/~rusin/known-math/collection/mathmap.html (visited on Dec. 2, 2012).

[Rus02] Dave Rusin. *Information about the Mathematical Atlas collection*. 2002. URL: http://www.math.niu.edu/~rusin/known-math/collection/index.html (visited on Dec. 2, 2012).

[Wik13] Wikipedia. *Zentralblatt MATH*. 2013. URL: http://en.wikipedia.org/wiki/Zentralblatt_MATH (visited on Feb. 26, 2014).

[Wik14a] Wikipedia. *D3.js*. 2014. URL: http://en.wikipedia.org/wiki/Protovis (visited on Jan. 25, 2014).

[Wik14b] Wikipedia. *Google Chart API*. 2014. URL: http://en.wikipedia.org/wiki/Google_Chart_API (visited on Jan. 25, 2014).

[Zen13a] Zentralblatt. *zbMATH*. 2013. URL: http://zbmath.org/?q=py:201* (visited on May 9, 2014).

[Zen13b] Zentralblatt. *zbMATH About*. 2013. URL: http://zbmath.org/about/ (visited on Feb. 26, 2014).

Appendix

Survey questions and results

In all questions 10 encoded the highest possible value while 1 was the lowest.

Question	Person 1	2	3	4	5	Mean
How simple was it to filter results on the map?	9	8	7	6	4	6.8
How simple was it to filter results in the zbmath search?	1	2	8	2	2	3
How simple was it to find the author you were looking for in the map?	7	8	5	9	6	7
How simple was it to find the author you were looking for in the zbmath search?	4	9	7	8	6	6.8
How simple was it to use the timeline tool?	9	9	9	8	9	8.8
How useful was the timeline tool?	9	10	10	10	8	9.4
Did you find the map to be useful to understand how results are related?	8	8	10	10	7	8.6
How likely is it that you would use the map for similar tasks again?	10	7	8	9	4	7.6
How helpful was the extra information provided about the author?	9	8	8	8	3	7.2